

3rd NF :- For a table to be in 3rd Normal form:

→ it should be in 2nd Normal form.

→ And it should not have Transitive Dependency

with exam-name and total-marks fields added the scoreTable look like this.

scoreid student-id subject-id marks exam-name totalmarks

Note. primarykey for scoretable is a compositekey
student-id + subject-id

a new column exam-name depends on the
Primarykey.

student-id + subject-id

total-marks depends on exam-name.

so the total marks depends on exam-name.
because each exam has own score.

$$E \rightarrow E + E$$

$$E \rightarrow E * E$$

$$E \rightarrow a$$

$$E \rightarrow b$$

So the primary key is studentid + subjectid
all attributes are depend on it except total marks.
which depend on exam-name.

but exam name is not the part of primary
key.

This is transitive D.

When there is an attribute in Table
which depend on non-prime attributes
and not on the prime attributes.

So solution is

exam-name total marks.
put these columns on another Exam Table.

student Table.

st-id name reg-no branch address.

Subj Table

subj-id subj-name teacher

Score Table

score-id student-id subject-id marks exam-name

Exam Table

exam_name total_marks

conclusion:-

if

A

→ B

non-prime
attribute

non-prime
attribute

Thus A known as FD.